EXPLORING *ALADIN* SKY ATLAS THROUGH SOUND

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ABSTRACT

Aladin is an interactive sky atlas that allows the visualization of images from astronomical catalogs and databases. It provides access to related data and information of all known astronomical objects through services such as *Simbad* (basic data, cross-identifications, bibliography and measurements for objects outside the solar system) and *VizieR* (standardized, published, and curated astronomical catalogs). As part of the Spanish Virtual Observatory (SVO) Open Science initiative, we have developed *SoniAladin*, an experimental Open Software voice-controlled interactive sonification approach to *Aladin*'s sky. This workshop aims to provide a horizontal introduction, both for Blind or Low Vision (BLV) and non BLV users, to the exploration of *Aladin* sky through sound. Participants will generate their own sonifications, starting from selected sky regions provided by the organizers, to finally freely explore Aladin Sky Atlas through sound. Each participant or group of participants will present their auditory results, including astronomical information of the represented objects, and any interesting detail of their sonification. All participants will provide feedback for each presentation using a brief online form in an open discussion.

1. WORKSHOP DESCRIPTION

Aladin Sky Atlas is a powerful interactive visualization tool for astronomical images, surveys, and catalogs, enabling users to define and explore specific sky regions. This workshop introduces the use of *SoniAladin* to transform Aladin's virtual sky into audible representations, making celestial patterns accessible through sound, and potentially providing a horizontal experience for both Blind or Low Vision (BLV) and non-BLV participants. Learning the basics of *Aladin*, we will capture sky regions to convert them into musical sequences, using a Python workflow and the built-in audio synthesizers included in Mac and Windows operating systems. The complete workshop is designed to last two hours. The first one will be dedicated to introduce the project, to provide assistance with the installation of the software (*Aladin* and *SoniAladin*), and to generate sonifications from sky regions provided by the organizers. During the second hour, participants will be able to generate customized sonifications from any region of the sky, freely exploring the sky atlas, and modifying any aspect of the sonification process based on their coding skills, sound design tools or preferences. No previous knowledge is required to participate in the workshop. At the end of the workshop, each participant or group of participants will showcase their best auditory results, including astronomical information of the represented objects and any interesting detail of their sonification process. The rest of participants will provide feedback for each presentation using a brief online form in an open discussion.

2. SOFTWARE INSTALLATION

Participants can find in the workshop github[1] all the required software installation instructions. It is recommended to have all the software pre-installed before the beginning of the session. Nevertheless, after the introduction, it will be possible to install *Aladin* and *SoniAladin* software with the assistance of the organizers. Additionally, some computers with all the software pre-installed will be available to ensure the participation. The proposal requires the following resources:

- A good internet connection to use *Aladin Lite* or the installation of *Aladin Desktop* for the visualization of the sky regions.
- The installation of *SoniAladin* for launching the MIDI notes and for supporting the user interface (UI), based on voice recognition.
- A built-in audio synthesizer or any user-selected virtual instrument or Digital Audio Workstation (DAW) to generate the sonifications.

2.1. *Aladin* Sky Atlas

In this workshop we will use the web app *Aladin Lite*. Access the app in the following link: `https://aladin.cds.unistra.fr/AladinLite/` Nevertheless, the following instructions allow the installation of *Aladin Desktop*, the free main application of the *Aladin* Sky Atlas suite, distributed under GPL3 license (requires a Java Virtual Machine to run).

1. If your system doesn't have a Java Virtual Machine installed, you can download and install it from the following link: `https://www.java.com/en/`. Just click on "Download Java for Desktops" and follow the instructions.
2. Download and install the *Aladin* Desktop version according to your operating system from the CDS webpage. `https://aladin.cds.unistra.fr/java/nph-aladin.pl?frame=downloading`

[1] ICAD 2026 Workshop Github: `https://github.com/AuditoryVO/ICAD2026_Workshop`

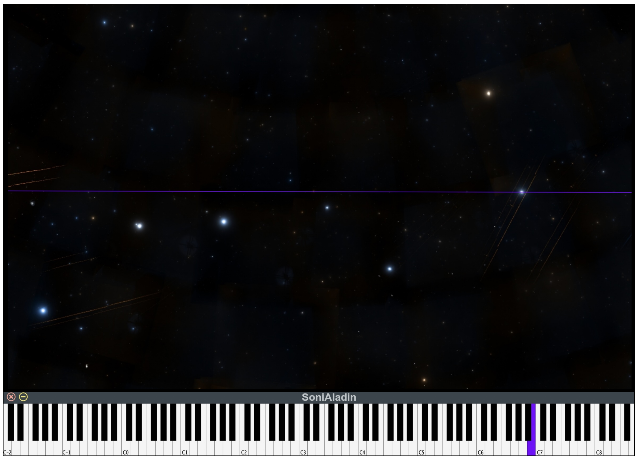

Figure 1: *SoniAladin* mapping description. The application explores *Aladin*'s screen captures from top to bottom. At each exploration step, bright objects launch the piano key notes with equivalent horizontal position (the lowest notes correspond to the bright objects on the left, and the highest notes correspond to the bright objects on the right). The amplitude of each note corresponds to the apparent luminance of the object. The default application provides user-controlled resolution, and luminance threshold, allowing the separation of the objects of interest from the background.

2.2. *SoniAladin* Installation

SoniAladin is an open software application that we have developed in the context of the Open science initiative of the Spanish Virtual Observatory (SVO) project. It is a Python module which launches MIDI notes related to the horizontal position of the bright objects appearing in voice-controlled screen captures. It was developed to be used in combination with *Aladin* Desktop to allow the sonification of its sky regions. The application captures the screen and explores the resulting image from top to bottom. As shown in Figure 1, on each exploration line, each bright object is converted into a note of the piano keyboard depending on its horizontal position (with the lowest notes corresponding to the bright objects on the left, and the highest notes corresponding to the bright objects on the right). The apparent luminance of each object controls the amplitude of the sounds generated by the built-in audio synthesizers included in Mac and Windows operating systems. The application aims at providing a first approach to the exploration of *Aladin* Sky Atlas through sound, potentially allowing a horizontal experience for both BLV and non-BLV participants.

In this workshop we will use the SoniAladin.py version provided in ICAD 2026 Workshop Github: `https://github.com/AuditoryVO/ICAD2026_Workshop`.

By default, *SoniAladin* picks the built-in MIDI output of the operating system. On Windows, that is the “Microsoft GS Wavetable Synth”. On macOs, the configuration of an IAC bus is required to route MIDI messages to a DAW such as “GarageBand”.

2.3. Only for Mac users. IAC configuration and GarageBand

This section is only for mac users. If you are working on Windows you should skip it completely. The following steps allow the configuration of the IAC Driver for routing Python's MIDI output to GarageBand. To enable and configure the IAC Driver:

1. Open Audio MIDI Setup (Applications / Utilities / Audio MIDI Setup).
2. In the menu bar choose Window Show MIDI Studio; a window with MIDI devices/icons appears.
3. Doubleclick the IAC Driver icon to open its properties.
4. In the properties window, check Device is online to enable the virtual MIDI device.
5. Close the IAC Driver properties and Audio MIDI Setup; the IAC bus is now available systemwide.

To set up GarageBand for listening to the IAC bus:

1. Launch GarageBand.
2. In GarageBand, open an ”Empty Project” and select ”MIDI” Software Instrument (it loads the default Classic Electric Piano).
3. Make sure that the track is selected/highlighted so it will receive MIDI input.
4. In GarageBand's track or project settings (depends on the version), ensure the MIDI input is set to receive from All MIDI Inputs (default) or specifically from the IAC Driver / Python bus if GarageBand exposes individual inputs.
5. Use the virtual keyboard to check if the piano is sounding. Open the virtual keyboard on Window/Show Keyboard and press any key. You should hear the piano.

2.4. Only for Windows users. General MIDI and Microsoft GS Wavetable Synth

This section is only for Windows users. If you are working on Mac you should skip it completely. On Windows, *SoniAladin* automatically picks Microsoft GS Wavetable Synth, and plays directly. GS Wavetable is fully builtin and reachable as a named MIDI output port, so Python MIDI output sounds with no extra configuration. Of course, Windows users can also use dedicated instruments or DAWs. Nevertheless, to change the GM Wavetable instruments, participants can easily modify *SoniAladin* code. General MIDI instruments are assigned to programs (0-127). The following example changes channel 1 from piano (0) to harp (46):

```
channel = 1
program_number = 46
status = 0xC0 + (channel - 1)
midi_out.send_message([status, program_number])
```

3. HANDS ON *ALADIN* SKY EXPLORATION THROUGH SOUND

Once all the participants have the system ready to use, we will explore the potential of the proposal learning the basics of *Aladin*, and generating sonifications of several curated sky regions. Table 1 provides a list of popular constellations to start exploring *Aladin*'s sky. IAU constellation maps [1] provide a useful resource to get oriented in regions with high density of bright objects. As shown in Figure 2, stars within constellations are designed with Greek letters which are needed to search in *Aladin* (try for instance “alf UMi” to go to *Polaris* star).

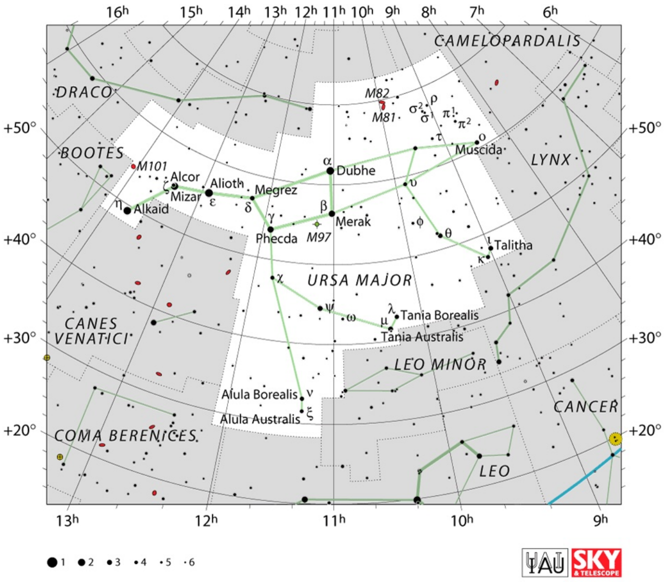

Figure 2: IAU sky chart for the Ursa Major constellation. The IAU charts were produced in collaboration with Sky & Telescope magazine (Roger Sinnott and Rick Fienberg). Alan MacRobert's constellation patterns, drawn in green on the charts, were influenced by those of H. A. Rey but, in many cases, were adjusted to preserve earlier traditions. Each chart was produced using the J2000 epoch. The images are released under the Creative Commons Attribution 4.0 International license.

4. CUSTOM PROJECT AND PRESENTATIONS

Aimed at generating customized sonifications from any region in the sky, participants will freely explore *Aladin* sky atlas, individually or in groups. No previous knowledge will be necessary to enjoy the workshop. Nevertheless, advanced participants will be able to modify any aspect of the sonification process based on their coding skills, sound design tools or preferences. Organizers will provide assistance with any potential issue, and guidance for any potential improvement proposed.

Participants with special interest in sound design are invited to explore the use of different virtual instruments and DAWs to generate their own sonifications. GarageBand provides a complete library of instruments and sounds, which allow the customization of the sonifications. On the other hand, Windows users can change the GM instruments with a simple edition of the *SoniAladin* code. Furthermore, any audio tool or DAW (such as MAX, Pure data, Supercollider, Ableton Live, Pro Tools, Logic or Reason), can receive the standard MIDI messages launched by *SoniAladin*, and could be used to generate complex soundscapes with the introduction of additional thresholds and instruments. This approach allows the creation of multi-layer luminance-based representations, which potentially help in the understanding of the auditory representation of images with a high density of bright objects. Participants with coding skills and special interest in sonification design are invited to implement any solution such as controlling the number of notes launched by one object, or reducing the overload of MIDI notes when lowering the apparent luminance thresholds.

At the end of the workshop, each participant or group of participants will present their best auditory results, including a brief description of the represented objects or sky regions, links to related publications, and any interesting detail of the sonification process, such as mapping descriptions, tools used, problems found or improvement proposals.

Table 1: List of 10 popular constellations with IAU abbreviations and English names [1]

Constellation	Abbreviation	English name
Ursa Major	UMa	The great bear
Ursa Minor	UMi	The little bear
Orion	Ori	The hunter
Cassiopeia	Cas	The Queen
Saggitarius	Sgr	The archer
Leo	Leo	The Lion
Taurus	Tau	The bull
Gemini	Gem	The twins
Andromeda	And	The princess
Pegasus	Peg	The winged horse

5. DISCUSSION AND FEEDBACK

After each presentation, all participants will have the opportunity to contribute in an open discussion aimed at providing feedback on relevant topics for sonification projects. Participants will additionally record their feedback through a brief online form, , available at: `https://forms.cloud.microsoft/e/WAGY3zhKqW` . The questionnaire aims at analyzing the usefulness, decision making, accuracy, and aesthetics of the representations. The anonymous feedback recorded is expected to be used in a journal publication.

6. ACKNOWLEDGMENT

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7. REFERENCES

[1] IAU, "Constellations reference," https://www.iau.org/IAU/Iau/Science/What-we-do/The-Constellations.aspx.

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[3] T. Boch and P. Fernique, "Aladin lite: Embed your sky in the browser," *Astronomical data analysis software and systems XXIII*, vol. 485, p. 277, 2014.

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